Do Now · retrieval

Recursion: base cases and returns

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A C# function is declared int Double(int n). What does return n * 2; do when n is 4?

Q2

Starting total = 1, repeat total = total * i for i = 1, 2, 3. What is total after each iteration?

Q3

Values A, B and C are pushed onto a stack in that order. Which leaves first when popping?

Q4

A function AddTwo(x) returns x + 2. Calculate AddTwo(AddTwo(3)) and show the inner and outer calls.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start with a small input — Factorial(4) calls Factorial(3); it cannot finish multiplying until the inner call returns.

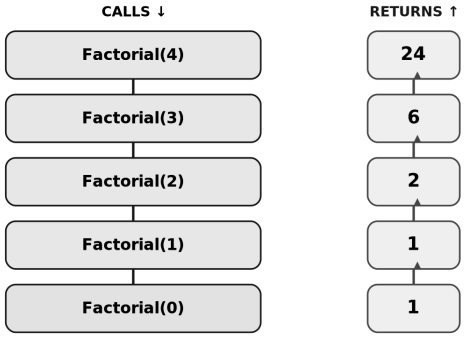
Core idea 2

Reach the stopping rule — Factorial(0) returns 1. It makes no further recursive call.

Core idea 3

Return through the calls — Factorial(1) = 1 × 1; Factorial(2) = 2 × 1; Factorial(3) = 3 × 2; Factorial(4) = 4 × 6 = 24.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

recursion — A function calling itself to solve a smaller problem.

base case — A condition that stops further recursive calls.

recursive case — A call to the same function with a smaller or simpler input.

call stack — Memory tracking active function calls and their local state.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

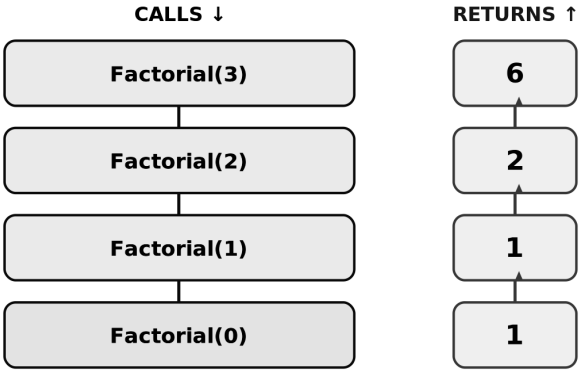
Problem

Trace Factorial(3). Show each call and its return value.

Follow the worked method

1. Calls — 3 → 2 → 1 → 0. The argument decreases by one each time.
2. Base case — Factorial(0) returns 1. No more calls occur.
3. Returns — Factorial(1)=1, Factorial(2)=2, Factorial(3)=6. Final answer: 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next call
4 calls 3; 3 calls 2; 2 calls 1; 1 calls 0.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Complete the calls and returns for Factorial(2).

Question 2

What happens if Factorial has no base case? Explain using the call stack.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Modify Factorial to reject negative input; write the C# guard and explain the reason.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If a recursive call reaches its base case, all previous calls have already returned.

Use key vocabulary from memory

1. Explain recursion in your own words.

2. Explain base case in your own words.

3. Explain recursive case in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.